

Hw8

1.

so we let $A = \{1, 2, 3, 4, 5, 6\}$ and let

$$h : \mathcal{P}(A) \rightarrow \mathbb{Z}, \quad h(X) = \sum_{x \in X} x.$$

1. the domain of h is $\mathcal{P}(A)$, the power set of A , and the codomain of h is $\mathbb{Z}$

2.

$$h(\{2, 5\}) = 2 + 5 = 7, \quad h(\{1, 3, 4\}) = 1 + 3 + 4 = 8$$

3. the preimage of 7 under h is

$$h^{-1}(7) = \{X \in \mathcal{P}(A) : h(X) = 7\}$$

the subsets of A whose elements add to 7 are

$$\{1, 6\}, \{2, 5\}, \{3, 4\}, \{1, 2, 4\}$$

therefore we got,

$$h^{-1}(7) = \{\{1, 6\}, \{2, 5\}, \{3, 4\}, \{1, 2, 4\}\}$$

4. since all elements of A are positive, the smallest possible sum is 0 (from $\emptyset$), and the largest possible sum is

$$1 + 2 + 3 + 4 + 5 + 6 = 21$$

actually every integer from 0 to 21 is a subset sum, so

$$\text{Im}(h) = \{0, 1, 2, 3, \dots, 21\}.$$

checking:

$$\begin{array}{llllll} 0 = h(\emptyset), & 1 = h(\{1\}), & 2 = h(\{2\}), & 3 = h(\{3\}), & 4 = h(\{4\}), & 5 = h(\{5\}), \\ 7 = h(\{2, 5\}), & 8 = h(\{3, 5\}), & 9 = h(\{4, 5\}), & 10 = h(\{4, 6\}), & 11 = h(\{5, 6\}), & 12 = h(\{1, 6\}), \\ 14 = h(\{3, 5, 6\}), & 15 = h(\{4, 5, 6\}), & 16 = h(\{1, 4, 5, 6\}), & 17 = h(\{2, 4, 5, 6\}), & 18 = h(\{3, 4, 5, 6\}), & 19 = h(\{1, 2, 5, 6\}), \\ 20 = h(\{2, 3, 4, 5, 6\}), & 21 = h(A). \end{array}$$

5. h isn't one to one and a counterexample is

$$h(\{1, 6\}) = 7 = h(\{2, 5\})$$

but $\{1, 6\} \neq \{2, 5\}$ so distinct inputs can have the same output, so h isn't injective

6. h is not onto as a function from $\mathcal{P}(A)$ to $\mathbb{Z}$ since we already have

$$\text{Im}(h) = \{0, 1, 2, \dots, 21\},$$

so many integers are not hit by h so for example, there is no subset $X \subseteq A$ where $h(X) = 22$, and there is no subset $X \subseteq A$ such that $h(X) = -1$ so then h is not surjective onto $\mathbb{Z}$

2.

1. so we have

$$f = \{(1, 2), (3, 4), (5, 6)\} \quad g = \{(2, 1), (4, 3), (6, 5)\}$$

then

$$(g \circ f)(1) = g(f(1)) = g(2) = 1, \quad (g \circ f)(3) = g(f(3)) = g(4) = 3, \quad (g \circ f)(5) = g(f(5)) = g(6) = 5$$

so

$$g \circ f = \{(1, 1), (3, 3), (5, 5)\}.$$

also

$$(f \circ g)(2) = f(g(2)) = f(1) = 2, \quad (f \circ g)(4) = f(g(4)) = f(3) = 4, \quad (f \circ g)(6) = f(g(6)) = f(5) = 6.$$

so we got

$$f \circ g = \{(2, 2), (4, 4), (6, 6)\}.$$

2. we let $f(x) = 2x + 1$ and $g(x) = 4x - 5$ for all $x \in \mathbb{Z}$. then

$$(g \circ f)(x) = g(2x + 1) = 4(2x + 1) - 5 = 8x - 1.$$

also we got

$$(f \circ g)(x) = f(4x - 5) = 2(4x - 5) + 1 = 8x - 9.$$

so that means

$$g \circ f(x) = 8x - 1, \quad f \circ g(x) = 8x - 9$$

3.

since

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(n) = 2n + 1,$$

and

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = 2x + 1$$

(a) f isn't onto and every value of the form $f(n) = 2n + 1$ is odd so no even integer lies in the image of f and for example, there is no $n \in \mathbb{Z}$ such that

$$f(n) = 2$$

so $2n + 1 = 2$ would imply $n = \frac{1}{2}$, which is not an integer so f is not onto $\mathbb{Z}$

(b) g is a bijection

Injective: let's say $g(x_1) = g(x_2)$ so

$$2x_1 + 1 = 2x_2 + 1.$$

and subtracting 1 and dividing by 2 gives $x_1 = x_2$ so g is one to one

Surjective: we have $y \in \mathbb{R}$ and we need some $x \in \mathbb{R}$ with $g(x) = y$ so:

$$2x + 1 = y \implies x = \frac{y - 1}{2}.$$

since $\frac{y-1}{2} \in \mathbb{R}$ this choice works so g is onto

since g is both injective and surjective, g is a bijection

(c) to find the inverse, we let $y = 2x + 1$ and solve for x :

$$x = \frac{y - 1}{2}.$$

so

$$g^{-1}(x) = \frac{x - 1}{2}$$

(d) we compute the two compositions:

$$(g^{-1} \circ g)(x) = g^{-1}(2x + 1) = \frac{(2x + 1) - 1}{2} = x \quad (x \in \mathbb{R}),$$

and

$$(g \circ g^{-1})(x) = g\left(\frac{x - 1}{2}\right) = 2\left(\frac{x - 1}{2}\right) + 1 = x \quad (x \in \mathbb{R}).$$

so both compositions are the identity function on $\mathbb{R}$

(e) in general, if a function is invertible, then composing it with its inverse gives an identity function so

$$g^{-1} \circ g = \text{id}_{\mathbb{R}} \quad \text{and} \quad g \circ g^{-1} = \text{id}_{\mathbb{R}}$$

that means each composition returns every input unchanged

4.

we have $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by

$$f(n) = \begin{cases} n + 1 & \text{if } n \equiv 0 \pmod{3}, \\ n - 1 & \text{if } n \equiv 1 \pmod{3}, \\ n & \text{if } n \equiv 2 \pmod{3} \end{cases}$$

1. f is one-to-one and we prove this by checking residue classes modulo 3

every integer is congruent to one of 0, 1, 2 (mod 3) so since we got:

- if $n \equiv 0 \pmod{3}$, then $f(n) = n + 1 \equiv 1 \pmod{3}$;
- if $n \equiv 1 \pmod{3}$, then $f(n) = n - 1 \equiv 0 \pmod{3}$;
- if $n \equiv 2 \pmod{3}$, then $f(n) = n \equiv 2 \pmod{3}$

so the output residue class tells us the input residue class is: an output congruent to 1 came from a number congruent to 0, an output congruent to 0 came from a number congruent to 1, and an output congruent to 2 came from a number congruent to 2

now if $f(a) = f(b)$ then we consider the residue class of this common value

- If $f(a) = f(b) \equiv 1 \pmod{3}$, then $a \equiv b \equiv 0 \pmod{3}$, so

$$a + 1 = f(a) = f(b) = b + 1,$$

so $a = b$

- If $f(a) = f(b) \equiv 0 \pmod{3}$, then $a \equiv b \equiv 1 \pmod{3}$, so

$$a - 1 = f(a) = f(b) = b - 1,$$

then $a = b$.

- if $f(a) = f(b) \equiv 2 \pmod{3}$, then $a \equiv b \equiv 2 \pmod{3}$, so

$$a = f(a) = f(b) = b,$$

hence $a = b$.

in all cases $a = b$ so f is injective

2. f is onto, so $y \in \mathbb{Z}$ means we must find $n \in \mathbb{Z}$ such that $f(n) = y$
there are three cases

- if $y \equiv 0 \pmod{3}$, let $n = y + 1$ then $n \equiv 1 \pmod{3}$, so

$$f(n) = n - 1 = (y + 1) - 1 = y$$

- if $y \equiv 1 \pmod{3}$, let $n = y - 1$ then $n \equiv 0 \pmod{3}$, so

$$f(n) = n + 1 = (y - 1) + 1 = y$$

- if $y \equiv 2 \pmod{3}$, let $n = y$ and then $n \equiv 2 \pmod{3}$, so

$$f(n) = n = y$$

So every integer has a preimage, so f is surjective

3. since f is both one-to-one and onto, f is a **bijection** its inverse we get it by reversing each rule:

$$f^{-1}(n) = \begin{cases} n + 1 & \text{if } n \equiv 0 \pmod{3}, \\ n - 1 & \text{if } n \equiv 1 \pmod{3}, \\ n & \text{if } n \equiv 2 \pmod{3}. \end{cases}$$

actually, $f^{-1} = f$ so this function is its own inverse

5.

first partition the set $\{2, 3, 4, \dots, 12\}$ into six groups:

$$\{2, 12\}, \{3, 11\}, \{4, 10\}, \{5, 9\}, \{6, 8\}, \{7\}$$

for the first five groups, the two numbers in the group add up to 14
then:

- **Pigeons:** the 7 selected numbers
- **Holes:** the 6 groups listed above

since 7 pigeons are placed into only 6 holes, the Pigeonhole Principle says that some hole must contain at least two of the selected numbers

the only hole that cant really contain two distinct selected numbers is $\{7\}$, because it contains just one number so the two selected numbers that land in the same hole must lie in one of the first five holes:

$$\{2, 12\}, \{3, 11\}, \{4, 10\}, \{5, 9\}, \{6, 8\}$$

any such pair sums to 14 so some two of the selected numbers add up to 14
so the statement is true